E 6.1

$f(x) = x_1^3 + x_2^3 - 18x_1^2 - 16x_2^2 - 84x_1$

$\therefore \frac{\partial f}{\partial x_1} = 3x_1^2 - 36x_1 - 84$,

$\frac{\partial f}{\partial x_2} = 3x_2^2 - 36x_2$

$\therefore \nabla f = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}\right) = (3x_1^2 - 36x_1 - 84, 3x_2^2 - 36x_2)$

When $\nabla f = 0$ (FONC)

$3x_1^2 - 36x_1 - 84 = 3(x_1^2 - 12x_1 - 28) = 3(x_1+2)(x_1-14) = 0$

$\therefore x_1 = -2$ or $x_1 = 14$

$3x_2^2 - 36x_2 = 3x_2(x_2 - 12) = 0$

$\therefore x_2 = 0$ or $x_2 = 12$

$\frac{\partial^2 f}{\partial x_1^2} = 6x_1 - 36$, $\frac{\partial^2 f}{\partial x_2^2} = 6x_2 - 36$, $\frac{\partial^2 f}{\partial x_1 x_2} = 0$, $\frac{\partial^2 f}{\partial x_2 x_1} = 0$

$\therefore \nabla^2 f(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 x_2} \\ \frac{\partial^2 f}{\partial x_2 x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} 6x_1 - 36 & 0 \\ 0 & 6x_2 - 36 \end{bmatrix}$

① When $x_1 = -2$, $x_2 = 0$

$\nabla^2 f(x) = \begin{bmatrix} -48 & 0 \\ 0 & -16 \end{bmatrix}$ $\therefore \det\left(\begin{bmatrix} -48 & 0 \\ 0 & -16 \end{bmatrix}\right) > 0$, $tr\left(\begin{bmatrix} -48 & 0 \\ 0 & -16 \end{bmatrix}\right) < 0$ $\Rightarrow \begin{bmatrix} -48 & 0 \\ 0 & -16 \end{bmatrix}$ is negative semidefinite.

$\therefore$ By SOSC, $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \end{bmatrix}$ is a strict local maximizer

② When $x_1 = 14$, $x_2 = 0$

$\nabla^2 f(x) = \begin{bmatrix} 48 & 0 \\ 0 & -36 \end{bmatrix}$ $\therefore \det\left(\begin{bmatrix} 48 & 0 \\ 0 & -16 \end{bmatrix}\right) < 0$ $\therefore \begin{bmatrix} 48 & 0 \\ 0 & -16 \end{bmatrix}$ is indefinite

$\therefore \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 14 \\ 0 \end{bmatrix}$ is a saddle point.

③ When $x_1 = -2$, $x_2 = 12$

$\nabla^2 f(x) = \begin{bmatrix} -48 & 0 \\ 0 & 36 \end{bmatrix}$ $\therefore \det\left(\begin{bmatrix} -48 & 0 \\ 0 & 36 \end{bmatrix}\right) < 0$ $\therefore \begin{bmatrix} -48 & 0 \\ 0 & 16 \end{bmatrix}$ is indefinite

$\therefore \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -2 \\ 12 \end{bmatrix}$ is a saddle point.

④ When $x_1 = 14$, $x_2 = 12$:

$\nabla^2 f(x) = \begin{bmatrix} 48 & 0 \\ 0 & 36 \end{bmatrix}$ $\therefore \det\left(\begin{bmatrix} 48 & 0 \\ 0 & 36 \end{bmatrix}\right) > 0$ $tr\left(\begin{bmatrix} 48 & 0 \\ 0 & 36 \end{bmatrix}\right) > 0$

$\therefore \begin{bmatrix} 48 & 0 \\ 0 & 36 \end{bmatrix}$ is positive definite.

$\therefore \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 14 \\ 12 \end{bmatrix}$ is a strict local minimizer

$\therefore$ a): Saddle point: $\begin{bmatrix} 14 \\ 0 \end{bmatrix}$ $\begin{bmatrix} -2 \\ 12 \end{bmatrix}$

b): Strict local minimizer: $\begin{bmatrix} 14 \\ 12 \end{bmatrix}$

c): Strict local maximizer: $\begin{bmatrix} -2 \\ 0 \end{bmatrix}$

E 6.2 minimize $x^T A x$
subject to $x^T x = 2$
$x_i = 1$

a) $L(x, \lambda, \mu) = x^T A x + \lambda(2 - x^T x) + \mu(1 - x_i)$

$\nabla_x L = 2Ax - 2\lambda x - \mu e_i$, where $e_i = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \end{bmatrix}$

$\therefore$ Main constraint:

$\nabla_x L = 2Ax - 2\lambda x - \mu e_i = 0$

Primal Feasibility: $x^T x = 2$, $x_i = 1$

Complementarity conditions: $\lambda(2 - x^T x) = 0$, $\mu(1 - x_i) = 0$

$\therefore$ The KKT conditions:

$$2Ax - 2\lambda x - \mu e_i = 0$$
$$x^T x = 2$$
$$x_i = 1$$

① For $i = 1$ $2a_1 - 2\lambda - \mu = 0$

$\therefore \mu = 2a_1 - 2\lambda$

② For $i \geq 2$:

$\because A = diag(a_1, \cdots, a_n)$

$\therefore$ For the i-th component

$2a_i x_i - 2\lambda x_i = 0$

$\therefore x_i(a_i - \lambda) = 0$

$\therefore$ For $i = 1$: $\mu = 2a_1 - 2\lambda$

For $i \neq 1$: $x_i = 0$ or $a_i - \lambda = 0$

$\because A = diag(a_1, \cdots, a_n)$ has distinct diagonal entries.

$\therefore \lambda = a_i$ for at most one i, as λ is a single value

① If $\lambda = a_k$, $2 \leq k \leq n$

$\therefore x_i = 0$ for $i \neq 1, k$, $x_1 = 1$

$\because x^T x = 2$ $\therefore x_k^2 = 1$ $\therefore x_k = \pm 1$

$\therefore \mu = 2a_1 - 2\lambda = 2(a_1 - a_k)$

② For $i \geq 2$, $x_i = 0$

$\therefore x^T x = x_1^2 = 1 \neq 2$

$\therefore$ Impossible.

$\therefore$ All KKT points: For all $k = 2, \cdots, n$ $x = [1\ 0\ \cdots\ x_k = \pm 1\ 0\ \cdots 0]^T$, $\lambda = a_k$ $\mu = 2a_1 - 2a_k$
$(x_1 = 1,\ x_i = 0$ for $i \neq 1, k,\ x_k = \pm 1)$

Total number of KKT points: $2(n-1)$

(c) For $x = [1\ 0\ \cdots\ x_k = \pm 1\ 0\ \cdots 0]^T$, $A = diag(a_1, \cdots, a_n)$

$x^T A x = x_1^2 a_1 + x_k^2 a_k = a_1 + a_k$

$\therefore$ To minimize $x^T A x$,

The optimal value of $x^T A x$ is $a_1 + \min_{i=2,\cdots,n} a_i$

E 6.3 maximize $4x_1+3x_2+6x_3$

s.t. $5x_1+2x_2+3x_3 \le 15$

$3x_1+4x_2+x_3 \le 9$

$x_1+x_2+x_3 \ge 0$

$\therefore g_1(x) = 5x_1+2x_2+3x_3-15 \le 0$

$g_2(x) = 3x_1+4x_2+x_3-9 \le 0$

$g_3(x) = -x_1-x_2-x_3 \le 0$

$\therefore \nabla g_1 = [5\ 2\ 3]^T$, $\nabla g_2 = [3\ 4\ 1]^T$ $\nabla g_3 = [1\ 1\ 1]^T$

$\because$ We want to maximize $4x_1+3x_2+6x_3$

$\Rightarrow$ minimize $-4x_1-3x_2-6x_3$

$\therefore f(x) = -4x_1-3x_2-6x_3$ $\therefore \nabla f = [-4\ -3\ -6]^T$

$\because L(x,\lambda) = f(x)+\lambda g(x) = f(x)+\lambda_1 g_1(x)+\lambda_2 g_2(x)+\lambda_3 g_3(x)$

$\therefore$ The kkt conditions :

$\therefore$ The main condition:

$-4+5\lambda_1+3\lambda_2+\lambda_3 = 0$

$-3+2\lambda_1+4\lambda_2-\lambda_3 = 0$

$-6+3\lambda_1+\lambda_2-\lambda_3 = 0$

Primal Feasibility :

$g_1(x) = 5x_1+2x_2+3x_3-15 \le 0$

$g_2(x) = 3x_1+4x_2+x_3-9 \le 0$

$g_3(x) = -x_1-x_2-x_3 \le 0$

Dual Feasibility:

$\lambda_1, \lambda_2, \lambda_3 \ge 0$

Complementarity :

$\lambda_1(5x_1+2x_2+3x_3-15) = 0$

$\lambda_2(3x_1+4x_2+x_3-9) = 0$

$\lambda_3(-x_1-x_2-x_3) = 0$

E 6.4 minimize $x_1e^{x_1}+(x_2-1)^2+x_3^2$

subject to $x_1+x_2 \le 4$,

$x_3-x_2^2 \ge 2$

$x_1, x_2, x_3 \ge 0$

$\therefore f(x) = x_1e^{x_1}+(x_2-1)^2+x_3^2$

$\therefore \nabla f(x) = \begin{bmatrix} e^{x_1}(x_1+1) \\ 2(x_2-1) \\ 2x_3 \end{bmatrix}$,

$g_1(x) = x_1+x_2-4 \le 0$, $g_2(x) = x_3^2-x_2+2 \le 0$, $g_3(x) = -x_1 \le 0$, $g_4(x) = -x_2 \le 0$, $g_5(x) = -x_3 \le 0$

$\therefore \nabla g_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ $\nabla g_2 = \begin{bmatrix} 0 \\ 2x_2 \\ -1 \end{bmatrix}$, $\nabla g_3 = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$, $\nabla g_4 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$ $\nabla g_5 = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$

$L(x,\lambda) = f(x)+\lambda_1 g_1(x)+\lambda_2 g_2(x)+\lambda_3 g_3(x)+\lambda_4 g_4(x)+\lambda_5 g_5(x)$

$\therefore \nabla L(x) = \nabla f(x)+\lambda_1\nabla g_1(x)+\lambda_2\nabla g_2(x)+\lambda_3\nabla g_3(x)+\lambda_4\nabla g_4(x)+\lambda_5\nabla g_5(x)$

Main condition:

$e^{x_1}(x_1+1)+\lambda_1-\lambda_3 = 0$

$2x_2-2+\lambda_1+2\lambda_2x_2-\lambda_4 = 0$

$2x_3-\lambda_2-\lambda_5 = 0$

Primal feasibility:

$x_1+x_2 \le 4$,

$x_3-x_2^2 \ge 2$

$x_1 \ge 0$

$x_2 \ge 0$

$x_3 \ge 0$

Dual Feasibility:

$\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5 \ge 0$

Complementarity:

$\lambda_1(x_1+x_2-4) = 0$

$\lambda_2(x_3-x_2^2+2) = 0$

$\lambda_3 x_1 = 0$

$\lambda_4 x_2 = 0$

$\lambda_5 x_3 = 0$

E 6.5. ① Proove SOSC $\Rightarrow$ OGC

$\therefore$ The point x^* satisfies $\begin{cases} \nabla f(x^*) = 0 \\ \nabla^2 f(x^*) \text{ is a positive definite matrix} \end{cases}$

According to the second-order Taylor expansion:

For $x \in B_\varepsilon(x^*)$, $\exists$

$f(x) = f(x^*)+\nabla f(x^*)(x-x^*)+\frac{1}{2}(x-x^*)^T\nabla^2 f(x^*)(x-x^*)+o(\|x-x^*\|^2)$

$\because \nabla f(x^*) = 0$,

$\therefore f(x)-f(x^*) = \frac{1}{2}(x-x^*)^T\nabla^2 f(\xi)(x-x^*)$, with ξ between x^* and x such that:

$\because \nabla^2 f(x^*)$ is positive definite, we have $\mu := \lambda_{min}(\nabla^2 f(x^*)) > 0$

$\because \xi \in B_\varepsilon(x^*)$ $\therefore \lambda_{min}(\nabla^2 f(\xi)) \ge \frac{1}{2}\lambda_{min}(\nabla^2 f(x^*)) > 0$, $\alpha = \frac{1}{2}\lambda_{min}(\nabla^2 f(x^*))$

$\therefore f(x)-f(x^*) = \frac{1}{2}(x-x^*)^T\nabla^2 f(\xi)(x-x^*) \ge \frac{1}{2}\alpha\|x-x^*\|^2$

$\therefore$ SOSC $\Rightarrow$ OGC.

② OGC $\Rightarrow$ SOSC :

$\because$ OGC

$\therefore f(x)-f(x^*) \ge \frac{\alpha}{2}\|x-x^*\|^2$, for all $x \in B_\varepsilon(x^*)$

For $x = x^*+td$, $\|d\| = 1$, $t \in (0,\varepsilon)$

$\therefore f(x^*+td)-f(x^*) \ge \frac{\alpha}{2}t^2$

According to the first-order derivative.

Suppose $\nabla f(x^*) \neq 0$

For $x = x^*+tv$, $v = -\nabla f(x^*)$:

$f(x^*+tv) = f(x^*)+t\nabla f(x^*)^T v+o(t)$

$\because \nabla f(x^*)^T v = -\|\nabla f(x^*)\|^2 < 0$

$\therefore f(x^*+tv) < f(x^*)$ $\Rightarrow$ contradiction

$\therefore \nabla f(x^*) = 0$

According to the second-order Taylor expansion

$f(x^*+td) = f(x^*)+t\nabla f(x^*)^T d+\frac{t^2}{2}d^T\nabla^2 f(x^*)d+o(t^2)$

$\therefore t\nabla f(x^*)^T d+\frac{t^2}{2}d^T\nabla^2 f(x^*)d+o(t^2) \ge \frac{\alpha}{2}t^2$

$\therefore \frac{\nabla f(x^*)^T d}{t}+\frac{1}{2}d^T\nabla^2 f(x^*)d+\frac{o(t^2)}{t^2} \ge \frac{\alpha}{2}t^2$

As $t \to 0^+$, $\frac{o(t^2)}{t^2} \to 0$

$\because \nabla f(x^*) = 0$

$\therefore d^T\nabla^2 f(x^*)d \ge 0$ for all d

$\therefore \nabla^2 f(x^*)$ is positive definite

$\therefore \nabla f(x^*) = 0$; $\nabla^2 f(x^*)$ is positive definite

$\therefore$ QGC $\Rightarrow$ SOSC

$\therefore$ Both SOSC $\Rightarrow$ QGC and QGC $\Rightarrow$ SOSC